

CAPSTONE PROJECT

Community Library Management System

A digital software system designed to streamline library operations — managing books, members, issuing, and returns in one organised platform. Reduces paperwork, minimises errors, and makes library services faster and more efficient.

TEAM MEMBERS: Sec 2

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ABSTRACT

- Our project The Community Library Management System is a software-based project designed to simplify and improve library operations. Traditional libraries often depend on manual registers and paperwork, which can be time-consuming and lead to errors. This system provides a digital platform to manage books, members, book issuing, returns, and overdue fines.
- It allows librarians to add, update, and search for books while helping members check book availability. The system also maintains borrowing records and calculates fines for late returns. By reducing paperwork, minimizing errors, and saving time, it makes library management more organized and efficient. The system can be developed using Java to provide a simple, reliable, and user-friendly library management solution.



INTRODUCTION TO COMMUNITY LIBRARY

What It Is
It is designed to manage library books, members, and daily library activities efficiently.

What It Does
It allows users to add and manage books and members, issue and return books, calculate fines, and view library records.

How It Works
The system is developed using Java concepts such as arrays, loops, conditions, switch statements, operators, and methods.

Where It Goes
Future improvements can include book search, sorting, multiple user accounts, automated notifications, and a graphical user interface (GUI).

Project Objectives

1

Manage Books

Add, update, and maintain book records efficiently

2

Manage Members

Store and manage library member details

3

Issue & Return Books

Keep track of issued and returned books

4

Calculate Fines

Calculate fines for late book returns

5

Maintain Records

Keep library information organized and reduce manual paperwork

Advantages

Easy Book Management

Makes it simple to add and manage book records

Quick Access to Information

Helps find book and member details easily

Saves Time

Reduces the time needed for manual record keeping

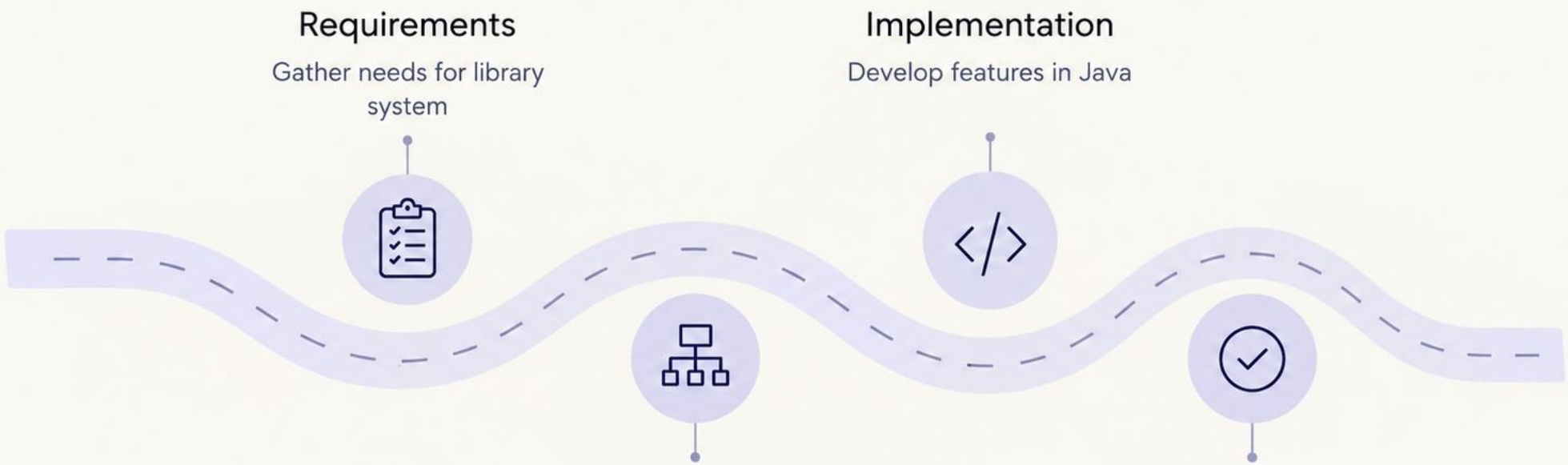
Reduces Errors

Minimizes mistakes in maintaining library records

Automatic Fine Calculation

Makes calculating late-return fines easier

Project Development Stages



Each stage builds on the previous — from identifying requirements and planning the system structure, through Java development and feature testing, to combining all modules

Course Outcomes

1. **Apply Programming Concepts**

   Use Java fundamentals such as variables, data types, operators, loops, and conditional statements

2. **Use Object-Oriented Concepts**

   Apply classes, objects, methods, and basic OOP principles

3. **Develop Problem-Solving Skills**

   Analyze a real-world library problem and develop a suitable software solution

4. **Implement Data Management**

   Store and manage book and member information using appropriate data structures

5. **Develop Modular Programs**

   Divide the project into separate functions or modules for easier development and maintenance

6. **Test and Debug Programs**

   Identify errors and test the system to ensure that its features work correctly

Topics Covered



Java Programming Basics

Variables, data types, operators, conditional statements, loops, switch statements, arrays

Object-Oriented Concepts

Classes, objects, methods, and basic OOP principles

Book and Member Management

Issue and return of books, fine calculation

Problem-Solving and Debugging

Identify errors and test the system

From Object-Oriented Design to Functional Java

Extending the Community Library Management System with core Java programming concepts

MODULE 4

OBJECT-ORIENTED MODELLING & EXCEPTION HANDLING

Classes and objects: fields, methods, new, references and null.

Constructors: default, parameterized, overloading and this.

Encapsulation: private fields, getters/setters, static members, constants and toString().

Inheritance, is-a relationships, interfaces and method overriding.

Exception handling: checked/unchecked exceptions, try-catch-finally, multi-catch, throw and throws.

MODULE 5

STRINGS, LIBRARY ALGORITHMS & FILE I/O

String operations: length, indexing, substring, concatenation, comparison and searching.

equals() vs ==; StringBuilder/StringBuffer; split(), join(), trimming and validation.

Searching and sorting with Arrays and Collections; Comparable and Comparator.

Read/write text files with Path, Files and buffered readers/writers.

Process records safely using try-with-resources; handle missing files and malformed data.

MODULE 6

COLLECTIONS, LAMBDAS & STREAMS

List, Set and Map using ArrayList, HashSet, HashMap and ordered collections.

Generics, type safety, autoboxing, for-each loops and iterators.

Functional interfaces and lambdas: Predicate, Function, Consumer, Supplier and Comparator.

Stream pipelines: filter, map, sorted, distinct, limit, forEach, collect, count and reduce.

Collectors and file streams: toList, groupingBy, counting, joining and reading lines.

PROJECT LINK: Model books and members as objects, store records in collections, and use files/streams for library reports.

IMPLEMENTATION & DEMO FLOW

Developed using Java with arrays, methods, loops, conditional statements, and switch statements. Features a **menu-driven interface** for different library operations.



Menu-driven interface keeps all library operations accessible from a single, intuitive console screen.

CONCLUSION

- The Community Library Management System provides an efficient, digital approach to managing books, members, and library transactions.
- It reduces manual workload, minimises errors, and makes library services faster, simpler, and more organised for everyone involved.

Future Scope

Future Enhancements

- 🔐 User Login Accounts
- 📧 Email & Notification Reminders
- 💳 Online Fine Payment
- 📱 Mobile Application
- 📲 QR / Barcode Book Management
- ☁️ Cloud-Based Database

✅ A practical solution for modern community libraries — built for simplicity and scale.

THANK YOU